

Transcil Android — Product & Software Ontology

Date: 2026-07-29

App: TranscilMobileApp (`com.transcil.rider`)

Audience: Anyone who needs a shared mental model — product, design, backend, or Android

Companion: Architecture deep-dive (class-level detail) lives beside this file

How to read this document

Every important idea appears in **three layers**:

Layer	Meaning
World	The real-world / business concept (what a rider or ops person would say)
App	How the Android client names and stores it
Wire	The gateway API or protocol that carries it

When layers disagree, **Wire (server) wins** for completion and verification. The app may cache drafts for UX, but the server is the authority.

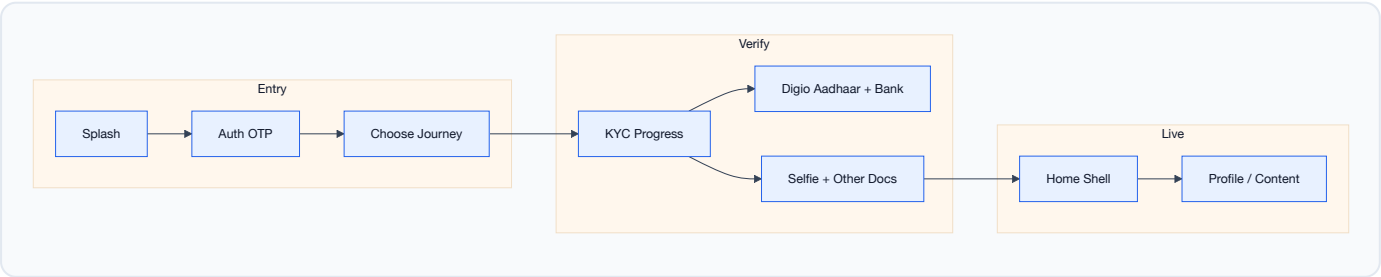
1. What this app is

Transcil’s Android **rider** app lets a person:

- 1. Sign in with a phone OTP
- 2. Choose a journey (**Rent EV** or **3PL**)
- 3. Complete **KYC** (identity + documents)
- 4. Enter a **Home** shell (hubs, battery swap, wallet, profile — some still local prototypes)

Identity / KYC traffic goes to the Transcil gateway (`api.transcil.in`). Digio runs in a browser tab; the phone never holds Digio secrets.

2. Big picture



Spine in one sentence: Splash decides where you are → Auth creates a session → Journey picks a role → KYC proves identity → Home is the post-verification product.

3. Core concepts (dictionary)

3.1 People & roles

Concept	World	App	Wire
Rider	The end user of this app	Profile drafts, home greeting	rider_id , GET/PATCH /v1/me/profile
Journey	Why they joined Transcil	JourneyType : RENT_EV / THREE_PL	rider_role : rider / 3pl via PUT /v1/me/rider-role
Session	“I’m logged in”	Encrypted tokens in TokenStore	access_token + refresh_token from /v1/auth/*

3.2 Identity & KYC

Concept	World	App	Wire
KYC	Know-your-customer checklist	kyc/* , KycStep , progress accordion	/v1/me/onboarding , /v1/kyc/* , /v1/me/kyc/*
Onboarding (API)	Server progress truth	OnboardingData → OnboardingSync.apply	GET /v1/me/onboarding
Onboarding (marketing)	Pre-login carousel	onboarding/OnboardingActivity	(none — local UI only)
Digio	Hosted Aadhaar + bank verify	Custom Tabs + DigioLauncher	POST /v1/kyc/start → gateway_url ; return transcil://kyc/callback
Aadhaar	National ID step	Step AADHAAR (Digio path)	Digio provider via KYC start/sync
Bank	Bank account step	Step BANK (Digio primary)	Digio; local IFSC draft may persist in prefs
PAN	Tax ID (3PL)	Step PAN	POST /v1/me/verify/pan
Selfie	Face capture for docs	SelfieVerification*	Upload-request → S3 PUT → submit
Other docs	Voter ID / DL / etc.	OTHER_DOCS , OtherDocsCatalog	Same upload pipeline; types from onboarding options.doc_types
Reference	Emergency contact (Rent EV)	Step REFERENCE	PUT /v1/me/reference
Documents status	Ops review of uploads	Drives Pending vs Approved	documents.overall , documents.verified

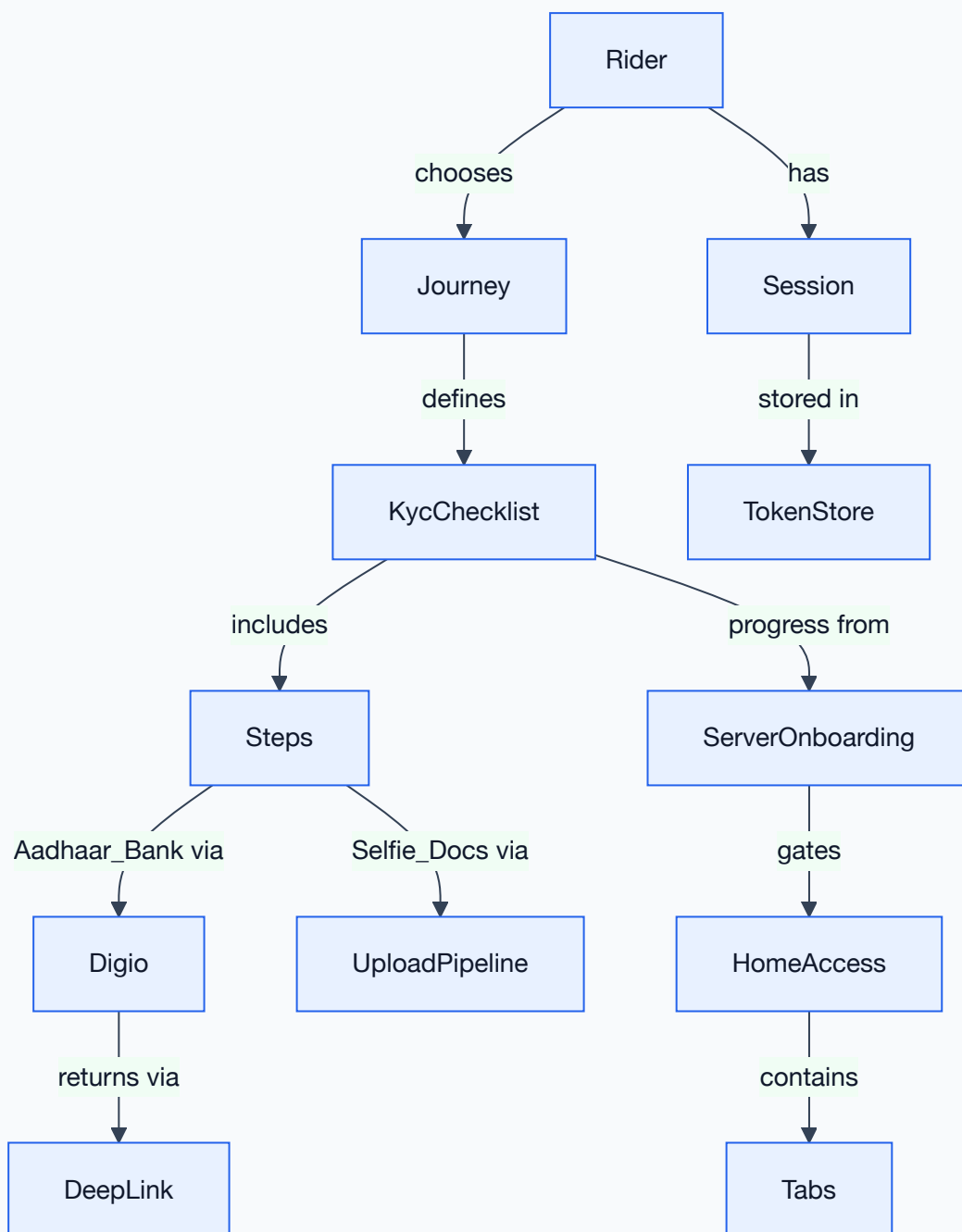
3.3 Product surfaces (post-KYC)

Concept	World	App	Wire
Home	Main app after KYC	<code>HomeDashboardActivity</code> + Nav tabs	Mostly local UI; profile logout is live
Nearby hub	Swap / service station	<code>NearbyHubs*</code> stub models	<i>(not API-backed yet)</i>
Battery swap	Swap history / actions	<code>BatterySwap*</code> stubs	<i>(not API-backed yet)</i>
Wallet	Balance & ledger	<code>Wallet*</code> stubs	<i>(not API-backed yet)</i>
Rental catalog	Vehicles & plans	<code>RentalCatalog</code> , Vehicles / Plans UI	<i>(local catalog — no booking API)</i>
Payment	Pay / autopay prototype	<code>PaymentActivity</code> step machine	<i>(UI only)</i>
Help / Terms / Privacy	Legal & support content	<code>ApiContent*</code> / <code>Help*</code>	GET <code>/v1/help-center</code> , <code>/v1/terms</code> , <code>/v1/privacy</code>

3.4 Software building blocks

Concept	World	App	Wire
Screen	A page the user sees	Activity or Fragment	—
ViewModel	Screen logic	<code>*ViewModel</code> , <code>BaseViewModel</code>	Calls repositories
Repository	Talks to the outside world	<code>AuthRepository</code> , <code>OnboardingRepository</code> , ...	Maps DTOs ↔ UI
API client	HTTP stack	<code>ApiClient</code> + <code>TranscilApi</code>	Retrofit/OkHttp/Gson
API envelope	Standard response shape	<code>ApiResponse<T></code>	<code>{ data, meta, error }</code>
Idempotency key	Safe retry of a write	UUID header on mutating calls	Idempotency-Key

4. Relationships (who owns what)



Ownership rules (memorize these):

1. **Server onboarding** owns “is this step done?”
2. **TokenStore** owns “is there a session?”
3. **KycProgressRepository** owns temporary form drafts for the accordion (not truth).
4. **DigioReturnSync** must refresh onboarding after Digio — never trust a local “green” alone.
5. **Documents.verified** owns entry to Approved Home.

5. KYC steps by journey

Same idea, different checklist order.

Step	Rent EV (rider)	3PL (3pl)	How it completes
Personal	Yes	Yes	PATCH /v1/me/profile
Address	Yes	Yes	PUT /v1/me/address
Aadhaar	Yes	Yes	Digio
Bank	Yes	Yes	Digio (primary)
Reference	Yes	Yes*	PUT /v1/me/reference
Other docs	Yes	Yes	Upload pipeline
PAN	—	Yes	POST /v1/me/verify/pan
Selfie	Yes	Yes	Upload pipeline

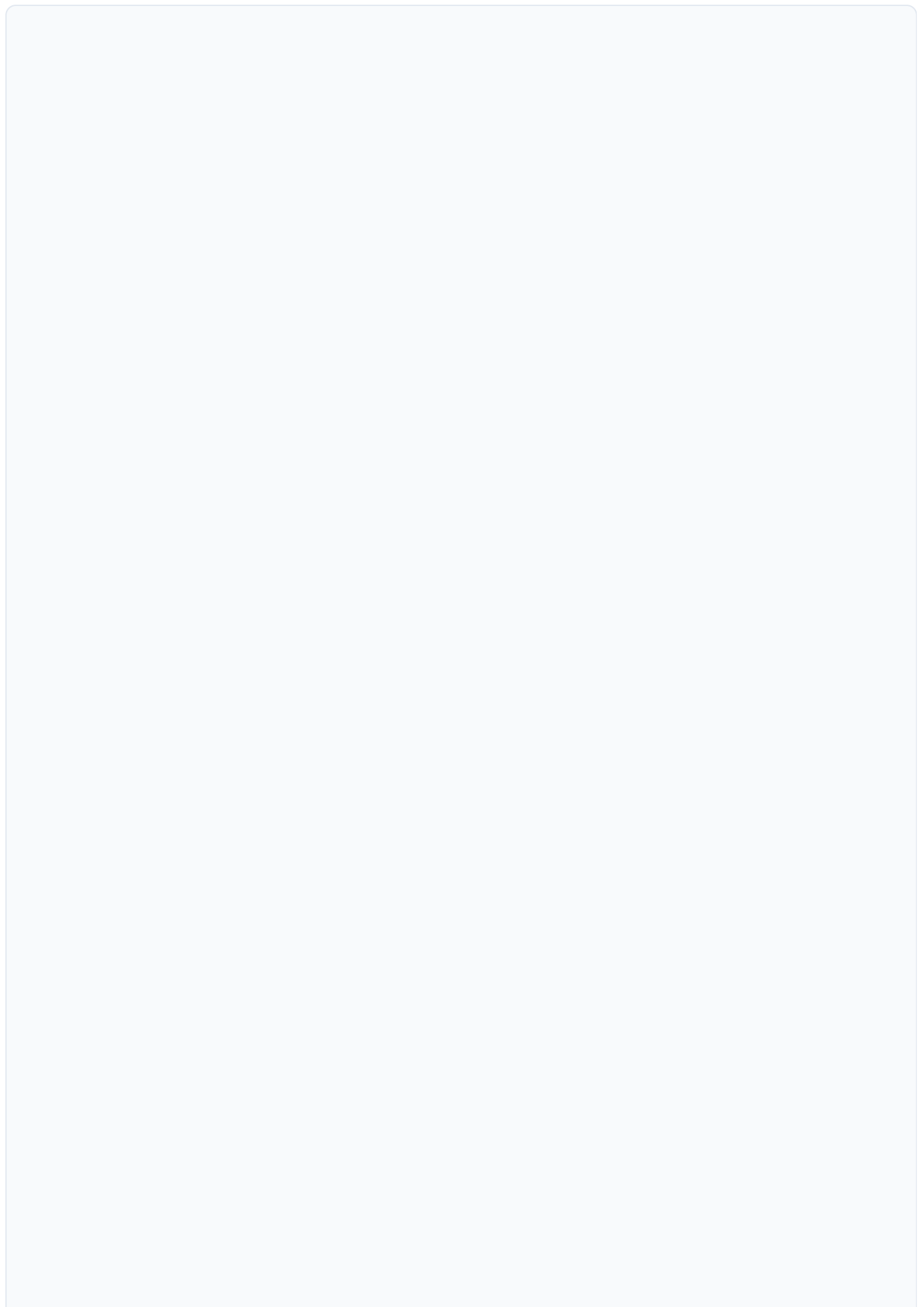
*UI catalog includes Reference for both; server `steps[]` is still the completion authority.

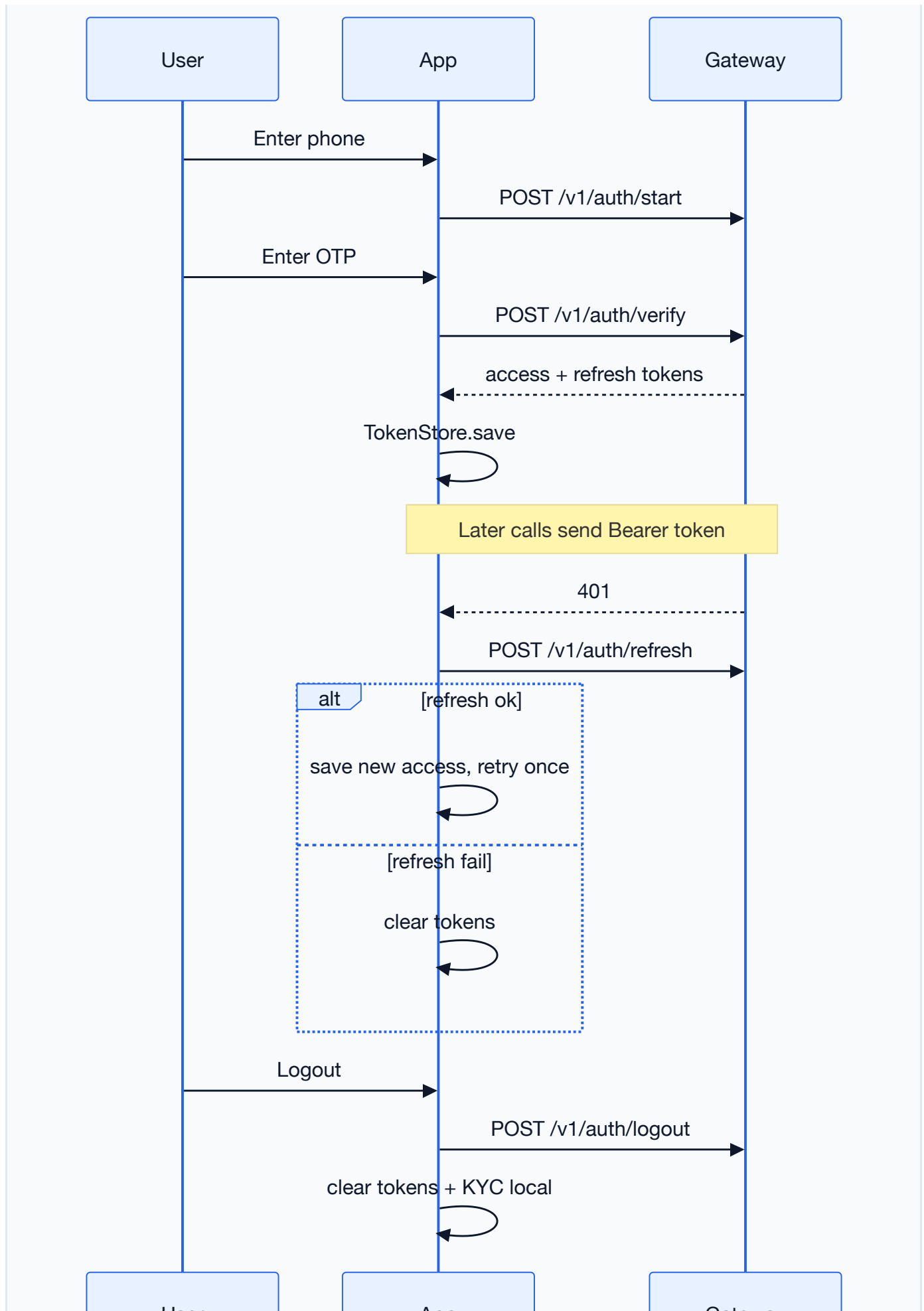
App enum: `KycStep` — `PERSONAL` , `ADDRESS` , `AADHAAR` , `BANK` , `REFERENCE` , `OTHER_DOCS` , `PAN` , `SELFIE` .

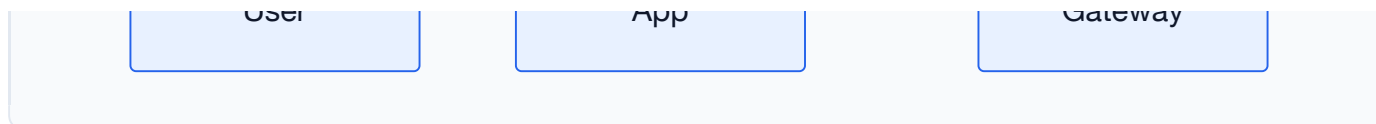
Local display order: `KycStepCatalog.stepsFor( journey )` .

6. Lifecycles

6.1 Session (login → logout)







6.2 Cold start (app opened again)

Condition	Destination
No token	Marketing onboarding → Welcome
Token, no rider_role	Choose Journey
Token, KYC incomplete	KYC Progress
Token, docs verified	Home (Approved)
Token, otherwise	Home (Pending)

Implemented in `AuthSession.resolveColdStartTarget`.

6.3 Digio return

1. App calls `POST /v1/kyc/start` with rider name
2. Opens `gateway_url` in Custom Tabs
3. Digio redirects to `transcil://kyc/callback`
4. `DigioReturnSync: sync-status` → `GET /me/onboarding` → `OnboardingSync.apply`
5. Accordion shows server status

6.4 Document upload

1. `POST /v1/me/kyc/upload-request` → presigned URL
2. Raw `PUT` bytes to S3/MinIO (with required headers + SHA-256)
3. `POST /v1/me/kyc/submit`
4. Refresh onboarding → Pending or Approved screen

7. Truth vs cache

Data	Authority	Survives process death?
Access / refresh tokens	<code>TokenStore</code> (encrypted prefs)	Yes
Step completion	Server <code>GET /v1/me/onboarding</code>	Yes (refetch)
Form drafts in memory	<code>KycProgressRepository</code>	No (rehydrate from server)
Bank IFSC local draft	<code>KycLocalStore</code> (plain prefs)	Yes (UX aid only)
Home hubs / wallet / swap	In-memory stubs	No — not real data

Rule of thumb: If you are deciding navigation or a green checkmark, ask the server. If you are filling a form the user typed a minute ago, the local draft is fine.

8. Where to look in the codebase

Concern	Package / entry
App start / splash	splash/MainActivity
Login / session routing	auth/*, especially AuthSession
Marketing carousel	onboarding/*
Rent EV vs 3PL	journey/*
KYC funnel	kyc/*
Home tabs	home/*
Vehicles / plans / pay UI	rental/*, payment/*
API + mapping	repository/*, data/network/*, data/model/*
Encrypted tokens / KYC prefs	data/local/*
Shared enums & UI helpers	core/*

Architecture pattern: Screen → ViewModel → Repository → TranscilApi (no DI framework; manual construction).

Navigation pattern: Intent stack for splash → auth → journey → KYC; Navigation Component only inside Home.

9. Status vocabulary

Term	Meaning
Step pending	Not started
Step in progress	Started, not finished
Step completed	Server says done
Documents in progress	Uploads under review
Documents verified	Ops approved — unlock Approved Home
Home PENDING	Can enter shell; KYC/docs not fully approved
Home APPROVED	documents.verified == true
Live API	Backed by gateway today
Stub / prototype	UI with local fake data — do not treat as production truth

10. Quick glossary

Say this	Means this
Transcil	EV rental / 3PL platform; this app is the rider client
Gateway	Backend API host (<code>api.transcil.in</code> or local <code>:4000</code>)
e164	Phone as <code>+91...</code>
OTP	One-time SMS code for login
Custom Tabs	Chrome tab used for Digio (safer than WebView)
Deep link	<code>transcil://kyc/callback</code> return into the app
OnboardingSync	Copies server progress into local drafts/UI
DigioReturnSync	Post-Digio sync + onboarding refresh
ApiClient	Singleton Retrofit/OkHttp setup
Bearer	<code>Authorization</code> header from access token
Phase A / B / C	Delivery slices: <code>auth</code> → Digio → <code>onboarding-as-truth</code>

11. What is deliberately out of scope (today)

- Real booking / payments settlement
- Live wallet money movement
- Live hub maps / battery inventory
- Digio SDK or Digio secrets on device
- Direct Aadhaar OTP happy path (superseded by Digio for Phase C)

Use the architecture deep-dive when you need class-by-class detail. Use **this ontology** when you need a shared language.

End of ontology — grounded in current Android code and Phase C identity contracts.